Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

56. Peer Skill Exchange & Mentorship Network

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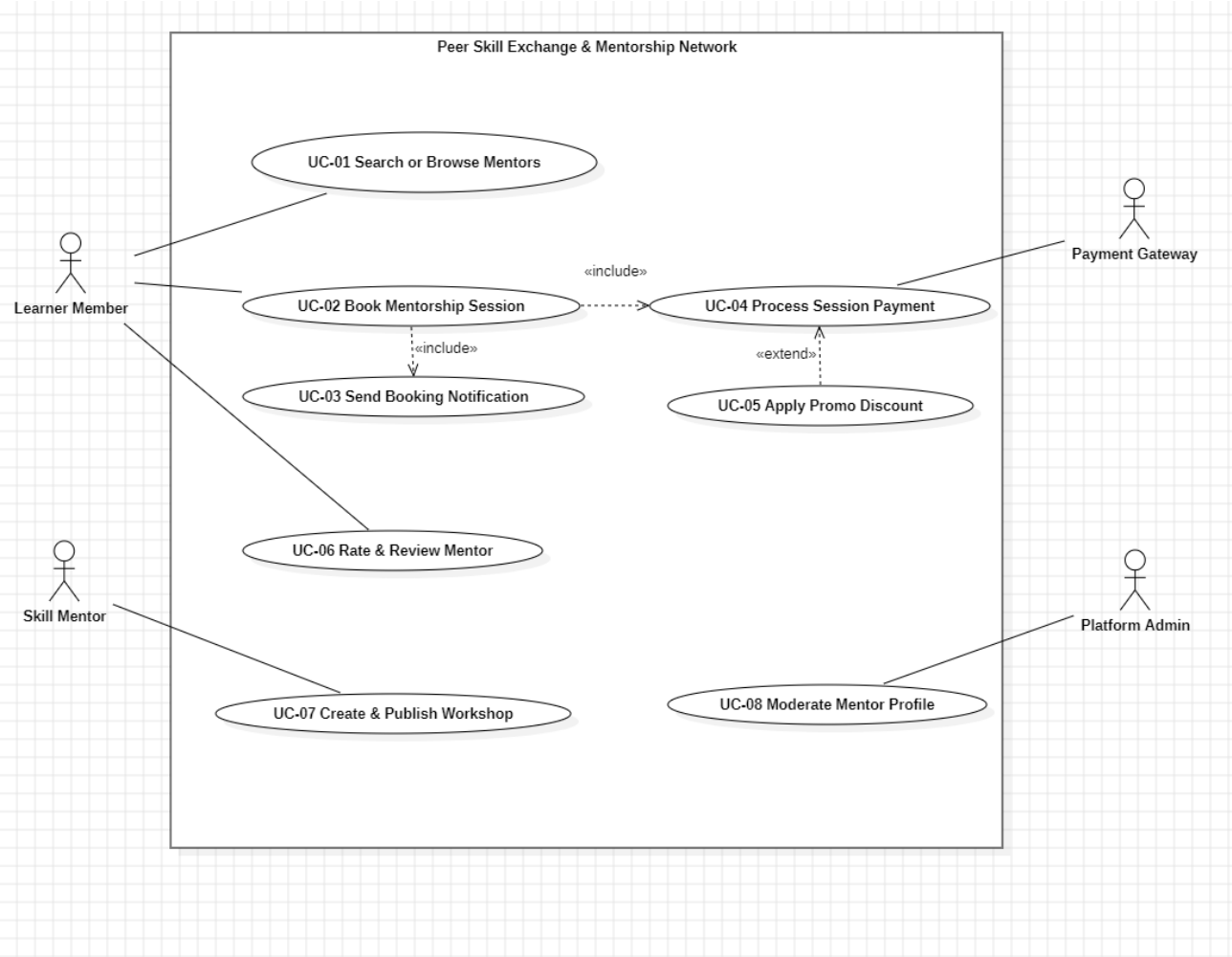
Deliverables :

1. Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Learner Member to search and filter Skill Mentors by skill category, availability, and rating.	High	Pass: When a learner enters a skill keyword and applies filters, results display only mentors matching all selected criteria within 2 seconds. Fail: Results contain mentors that do not match the selected criteria or take more than 2 seconds.	Core discovery function required for finding suitable mentors.
FR-002	Functional	The system shall allow a Learner Member to request and book a mentorship session with a Skill Mentor for a specific date and time slot.	High	Pass: When a learner selects an available slot and submits a booking request, the slot is marked Pending and the mentor receives a notification within 30 seconds. Fail: An unavailable slot is booked or the mentor is not notified within 30 seconds.	Booking is the primary interaction between learners and mentors.
FR-003	Functional	The system shall process payment for a booked mentorship session through an integrated payment gateway and support an optional promo code.	High	Pass: When payment is authorized, the booking status changes to Confirmed; when payment is declined, the booking remains Pending and the learner is notified. Fail: A declined payment results in a Confirmed booking.	Enables secure payment processing and supports the platform's paid mentorship model.

FR-004	Functional	The system shall allow a Learner Member to submit a star rating from 1 to 5 and a written review for a Skill Mentor within 7 days of a completed session.	Medium	Pass: After a session is completed, a review prompt is available for 7 days and a submitted rating/review appears on the mentor's profile within 1 minute. Fail: A learner can submit a review after the 7-day period or the review is not displayed within 1 minute.	Builds trust and provides feedback about mentorship quality.
FR-005	Functional	The system shall allow a Skill Mentor to create and publish a group skill-sharing workshop with a title, description, capacity, and schedule.	Medium	Pass: When a mentor publishes a workshop with valid details, it appears in the community events listing and accepts RSVPs up to the stated capacity. Fail: An invalid workshop is published or RSVPs exceed the specified capacity.	Extends the platform from individual mentorship to community-based learning.
NFR-001	Nonfunctional – Performance	The mentor search and filter results shall load within 2 seconds for a catalogue of up to 10,000 active mentor profiles.	High	Pass: Load testing shows the 95th-percentile search response time is below 2 seconds with 10,000 profiles. Fail: The 95th-percentile response time is 2 seconds or more.	Fast search is important for a smooth user experience.
NFR-002	Nonfunctional – Security	The system shall encrypt payment data and private learner-mentor messages in transit using TLS 1.2+ and at rest using AES-256, while supporting applicable data-subject rights.	High	Pass: Security testing confirms TLS 1.2+ for relevant endpoints, AES-256 for stored sensitive data, and processing of a valid data-deletion request within 30 days. Fail: Any required security control is absent or the deletion request exceeds 30 days.	Protects sensitive payment and personal information.

2. UML Use-Case Diagram



3. Use-Case Flow Specification:

Use Case: UC-02 - Book Mentorship Session

Primary Actor: Learner Member

Supporting Actor: Skill Mentor, Payment Gateway

Preconditions

- The Learner Member is logged into the platform with a verified account.
- The target Skill Mentor's profile is active and has at least one open availability slot.
- The learner has selected a mentor via UC-01 (Search / Browse Mentors).

Postconditions

- Success: The session is marked Confirmed, added to both the learner's and mentor's calendars, payment is captured, and a confirmation notification is sent to both parties.
- Failure: The session remains Pending or is Cancelled, and no payment is captured.

Main Success Scenario

1. Learner Member selects an available time slot on the Skill Mentor's profile and clicks “Book Session.”

2. System displays a booking form pre-filled with the mentor, date, time, and session duration.

3. Learner confirms the session topic and submits the booking request.

4. System validates the slot is still available and marks it Pending, then notifies the Skill Mentor.

5. Skill Mentor accepts the booking request within the platform.

6. System invokes UC-04 (Process Session Payment) — «include» — to charge the learner for the session, and UC-03 (Send Booking Notification) once the slot is marked pending.

7. Payment gateway authorizes the transaction.

8. System updates the booking status to Confirmed and adds the session to both calendars.

9. System sends a confirmation notification (email and in-app) with session details to the Learner Member and Skill Mentor.

10. Use case ends successfully.

Alternate Flow - 7a. Payment Declined

- 7a1. System displays a “Payment Failed” message to the Learner Member.
- 7a2. Learner Member is prompted to retry with the same or a different payment method.
- 7a3. If payment fails again after 2 attempts, the booking is automatically Cancelled, the slot is released back to the mentor's calendar, and both parties are notified.